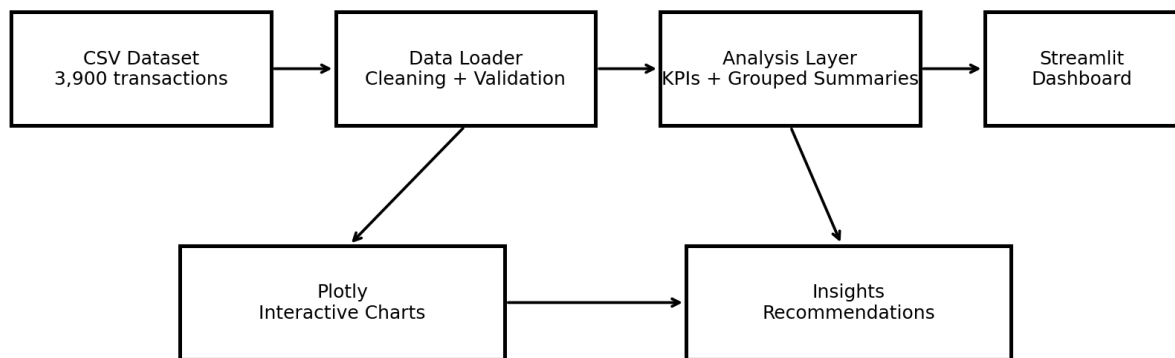


CUSTOMER SHOPPING BEHAVIOUR ANALYTICS

Interactive Streamlit Dashboard — Project Documentation

Customer Shopping Behaviour Analytics — System Architecture



Project objective: Transform customer transaction data into an interactive analytics dashboard that helps users explore sales, products, demographics, seasonality, purchasing behaviour and data-driven business recommendations.

Technology: Python • Pandas • Streamlit • Plotly • CSV

Dataset: 3,900 retail customer transactions

1. Introduction and Project Overview

Retail businesses collect large amounts of transactional information, but raw CSV data is difficult to interpret directly. This project converts customer shopping records into an interactive web-based analytics application. The dashboard is designed to provide a single place for exploring key performance indicators, customer segments, product performance and behavioural patterns.

Key Objectives

Objective	Description
Data preparation	Load, clean, validate and standardise the raw CSV data.
Performance analysis	Calculate revenue, purchase count, average spend, ratings and subscription metrics.
Customer analysis	Study gender, age groups, locations and subscriber behaviour.
Product analysis	Compare categories, products, purchase amounts and numeric relationships.
Behaviour analysis	Explore seasons, purchase frequency, discounts, promotions and repeat purchasing.
Decision support	Turn computed findings into practical business recommendations.

Application Pages

Overview & Data Quality — row count, column types, missing values and descriptive statistics.

Sales KPIs — total revenue, purchase count, average spend, average rating and subscription rate.

Product Analysis — top items, purchase amount distribution and correlation heatmap.

Demographics — gender, age, locations and subscriber breakdown.

Seasonal & Behavioural — seasonality, purchase frequency, discount impact and top customers.

Insights & Recommendations — dynamically calculated findings and actionable recommendations.

2. Dataset and Data Dictionary

The dashboard works from a single customer_shopping_behavior.csv file containing 3,900 retail customer transactions. Each record represents a customer purchase and contains demographic, product, transaction, promotion and payment information.

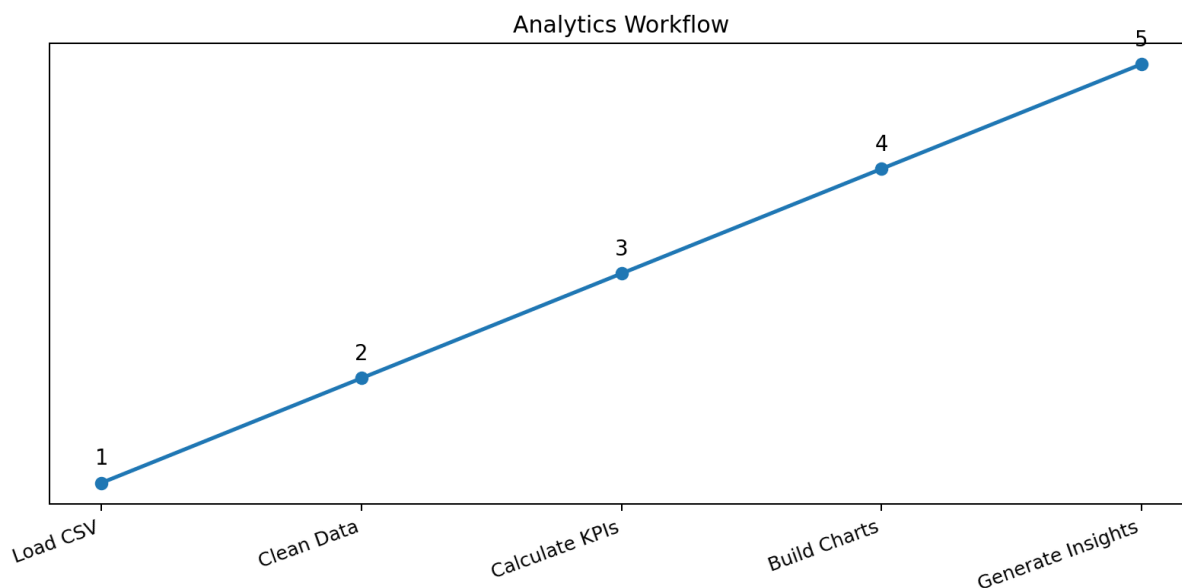
Column	Purpose
Customer ID	Unique customer identifier
Age / Gender	Customer demographic information
Item Purchased / Category	Product and product group
Purchase Amount (USD)	Transaction value
Location	US state/location
Size / Color	Product attributes
Season	Spring, Summer, Fall or Winter
Review Rating	Customer rating from 1.0 to 5.0
Subscription Status	Subscriber or non-subscriber
Shipping Type	Delivery/shipping option
Discount Applied / Promo Code Used	Promotion indicators
Previous Purchases	Count of prior transactions
Payment Method	Payment channel
Frequency of Purchases	Purchase frequency category

Important: The report does not hard-code business findings. The application calculates its insights dynamically from the cleaned dataset, so results change when the underlying data changes.

3. Data Cleaning and Processing

Reliable analytics depend on consistent input data. The data-loading layer therefore performs a sequence of validation and transformation operations before analysis begins.

Step	Processing performed	Purpose
1	Read CSV with utf-8-sig	Handles BOM characters in headers.
2	Rename columns to snake_case	Provides predictable Python-friendly field names.
3	Remove exact duplicates	Prevents duplicated transactions from distorting metrics.
4	Drop missing critical values	Keeps essential analytical records usable.
5	Coerce numeric fields	Ensures age, amount, rating and previous purchases are numeric.
6	Validate business values	Removes negative amounts and ratings outside 1–5.
7	Standardise categories	Strips whitespace and normalises categorical text.
8	Create boolean flags	Adds subscriber, discount and promo indicators.
9	Create age groups	Converts continuous age into interpretable segments.



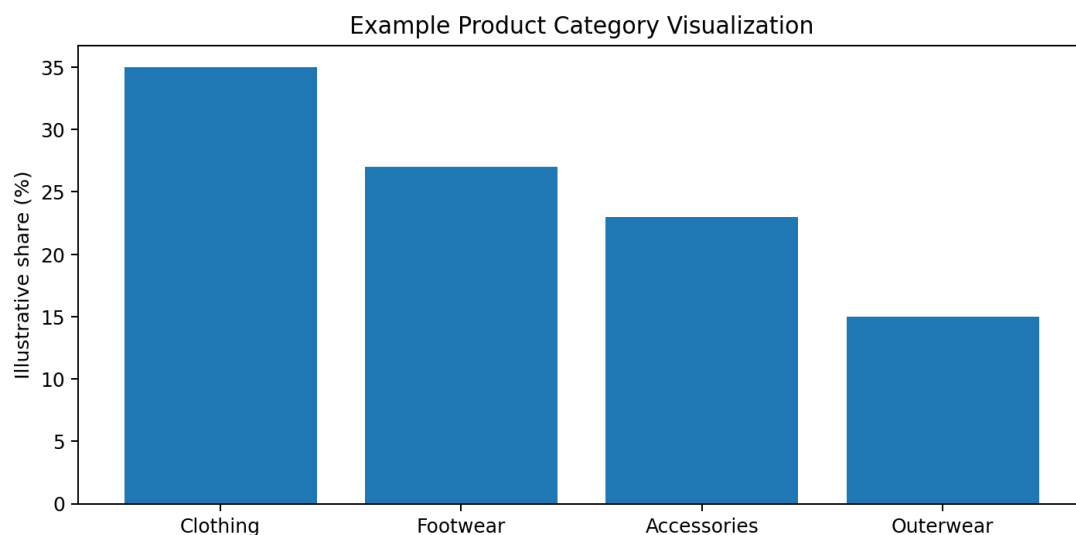
This processing pipeline separates raw-data concerns from the dashboard presentation layer, making the application easier to maintain and reducing the chance that charts operate on inconsistent values.

4. Analytics, KPIs and Visualisation

The analysis layer converts the cleaned DataFrame into summary metrics and grouped tables. Plotly is then used to present these results as interactive visualisations inside Streamlit.

Core KPIs

KPI	Meaning / calculation
Total Revenue	Sum of valid purchase amounts.
Purchase Count	Number of transaction records after cleaning.
Average Spend	Mean purchase amount per transaction.
Average Rating	Mean customer review rating.
Subscription Rate	Percentage of transactions/customers marked as subscribers, according to dashboard logic.



Illustrative visualization for documentation; not a reported dataset result.

Visualisations included in the application

Bar charts support product and location comparisons; pie/donut-style charts can show categorical splits; histograms show purchase amount distributions; scatter/correlation views help investigate relationships among numeric variables; and seasonal/frequency charts support behavioural analysis. Dashboard filters for Season, Gender and Category are intended to update the displayed analysis globally.

5. Dashboard Features and Business Insights

Interactive Dashboard Features

Feature	How it helps users
Global filters	Filter by Season, Gender and Category and inspect the selected segment.
Data quality view	Understand row counts, types, missing values and descriptive statistics.
Interactive Plotly charts	Hover, zoom and inspect chart values directly.
Top-product analysis	Identify products/items contributing strongly to transaction value.
Customer segmentation	Compare age groups, gender, location and subscription status.
Behaviour analysis	Study frequency, discounts, promotions and repeat purchases.
Dynamic recommendations	Generate recommendations from computed dataset patterns rather than fixed numbers.

Examples of Actionable Questions

- Which product categories generate the largest share of revenue?
- How does purchasing behaviour vary across seasons?
- What customer segments have higher purchasing frequency?
- How are discounts and promotional codes associated with purchase amounts?
- Which locations contribute strongly to sales?
- Do subscribers display different purchasing patterns from non-subscribers?
- Which customers have high previous-purchase counts and transaction activity?

These are analytical questions rather than pre-decided conclusions. The dashboard is designed to let users inspect the evidence through filters, tables and charts.

6. Technical Implementation, Deployment and Conclusion

Project Structure

app.py — Streamlit entry point and page/navigation logic.

src/data_loader.py — CSV loading and cleaning pipeline.

src/analysis.py — KPI calculations and grouped analytical summaries.

src/charts.py — Plotly chart helper functions.

data/customer_shopping_behavior.csv — source transaction dataset.

requirements.txt — Python package dependencies.

Execution

A recommended setup is to create a Python virtual environment, install the packages from requirements.txt, and run **streamlit run app.py**. The dashboard is designed to open locally at port 8501. A deployed Streamlit version is also referenced in the project README.

Technology Stack

Technology	Role
Python	Core programming and data-processing language
Pandas	Data loading, cleaning, transformation and aggregation
Streamlit	Interactive dashboard and web application layer
Plotly	Interactive visualisation
CSV	Lightweight source-data format

Conclusion

The Customer Shopping Behaviour Analytics Dashboard demonstrates an end-to-end data analytics workflow: raw transaction data is cleaned, transformed, analysed and communicated through interactive visualisations. Its modular structure separates data loading, analytical logic and chart generation, while Streamlit makes the results accessible through a simple browser interface. The project can be extended with predictive modelling, customer segmentation, sales forecasting, recommendation systems or database-backed data pipelines in future versions.

Project outcome: A reusable analytics dashboard for exploring 3,900 retail customer transactions and converting them into understandable, interactive business intelligence.

— End of Project Documentation —